

BOCCONI UNIVERSITY · COMPUTER VISION & IMAGE PROCESSING

Feedback-Augmented RT-DETR

A Cross-Attention Refinement Strategy for Small-Object Detection

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Causal contribution of feedback on COCO val2017:

+0.99

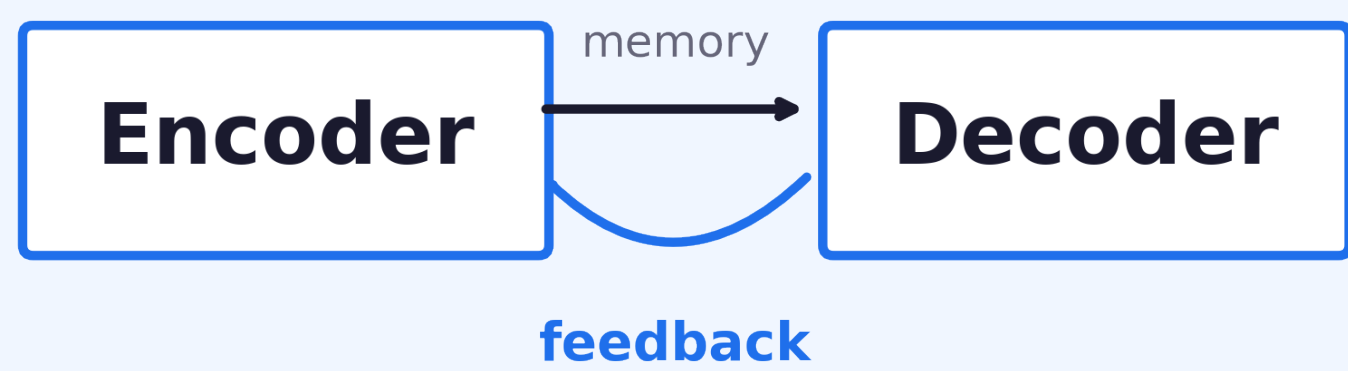
small-object AP (33.26 $\rightarrow$ 34.25, same checkpoint, ON – OFF)

1. The Problem

RT-DETR achieves real-time, NMS-free object detection. Yet on COCO val2017, RT-DETR-R50 reports $\text{AP}_S = 34.7$ for small objects versus $\text{AP}_L = 67.7$ for large ones — a 33-point gap on the same network.

Why? Encoder memory is computed once before any decoder layer fires. As the decoder discovers object hypotheses, that information cannot flow back to revise the features it is reading from. Refinement is strictly one-way.

2. Our Idea



After the first decoder layer makes preliminary predictions, use them as keys/values in a single cross-attention pass that refines the encoder memory. Layers 2– L then attend over the refined memory:

$$\mathbf{m}' = \mathbf{m} + g \cdot \text{Attn}(\mathbf{m}, \mathbf{t}^{(1)})$$

where $g \in (0, 1)$ is a learnable scalar gate.

3. v1: Trained but Silent

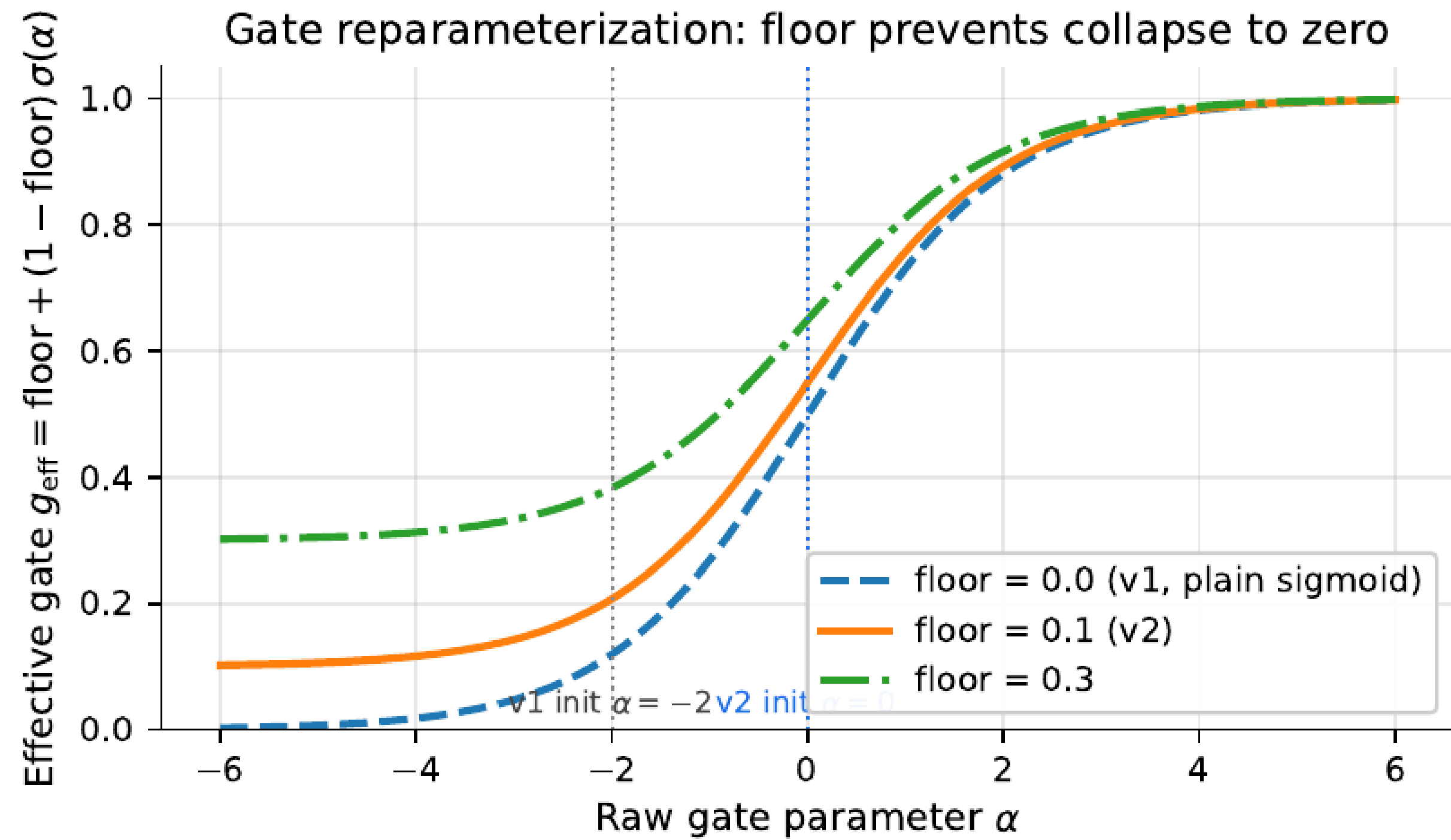
We added the module with a plain sigmoid gate ($g = \sigma(\alpha)$, $\alpha_{\text{init}} = -2$) applied to all four pyramid levels. Training was clean. The same-checkpoint inference ablation gave:

$\Delta \text{AP}_S = 0.00$

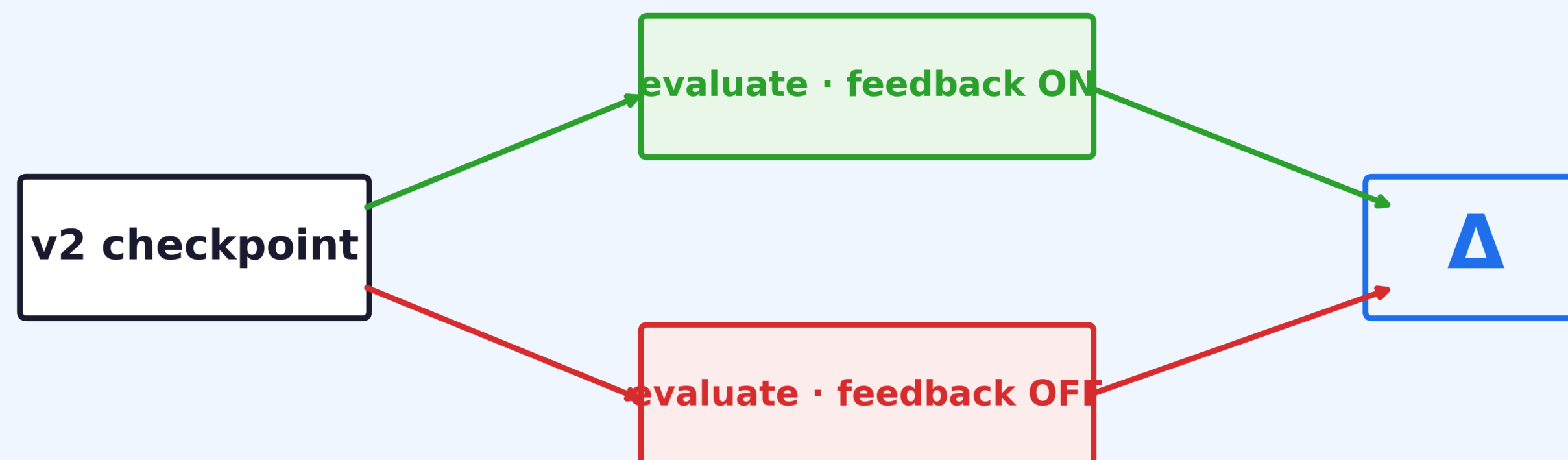
The gate had decayed to ≈ 0.12 during training. The cross-attention output was multiplied by ≈ 0 before reaching memory — statistically identical to skipping the module.

4. v2: Two Structural Fixes (Zero New Params)

- Gate floor reparameterization.**
 $g_{\text{eff}} = \text{floor} + (1 - \text{floor}) \cdot \sigma(\alpha)$, floor = 0.1.
The optimizer can attenuate, never silence.
- Level mask: P2/P3 only.**
Refine the stride-4 and stride-8 features (where small objects live); leave P4, P5 untouched.



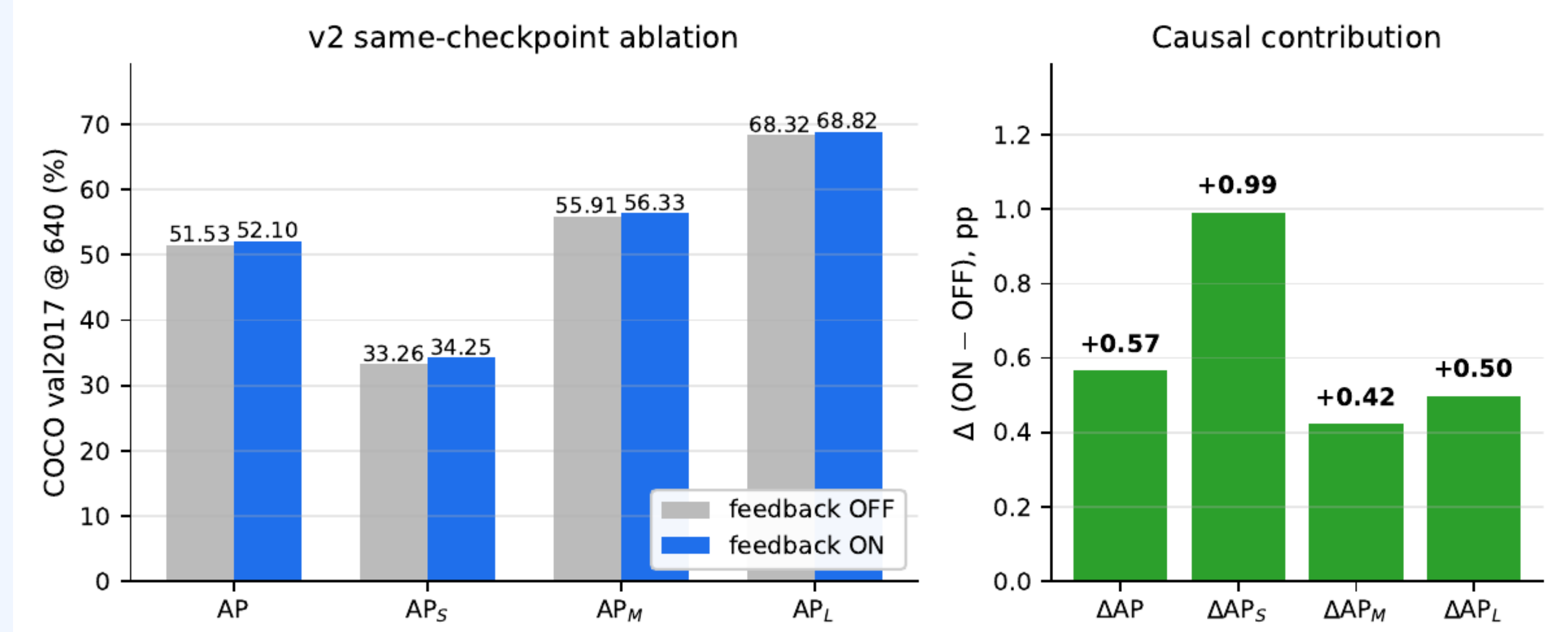
5. Same-Checkpoint Ablation



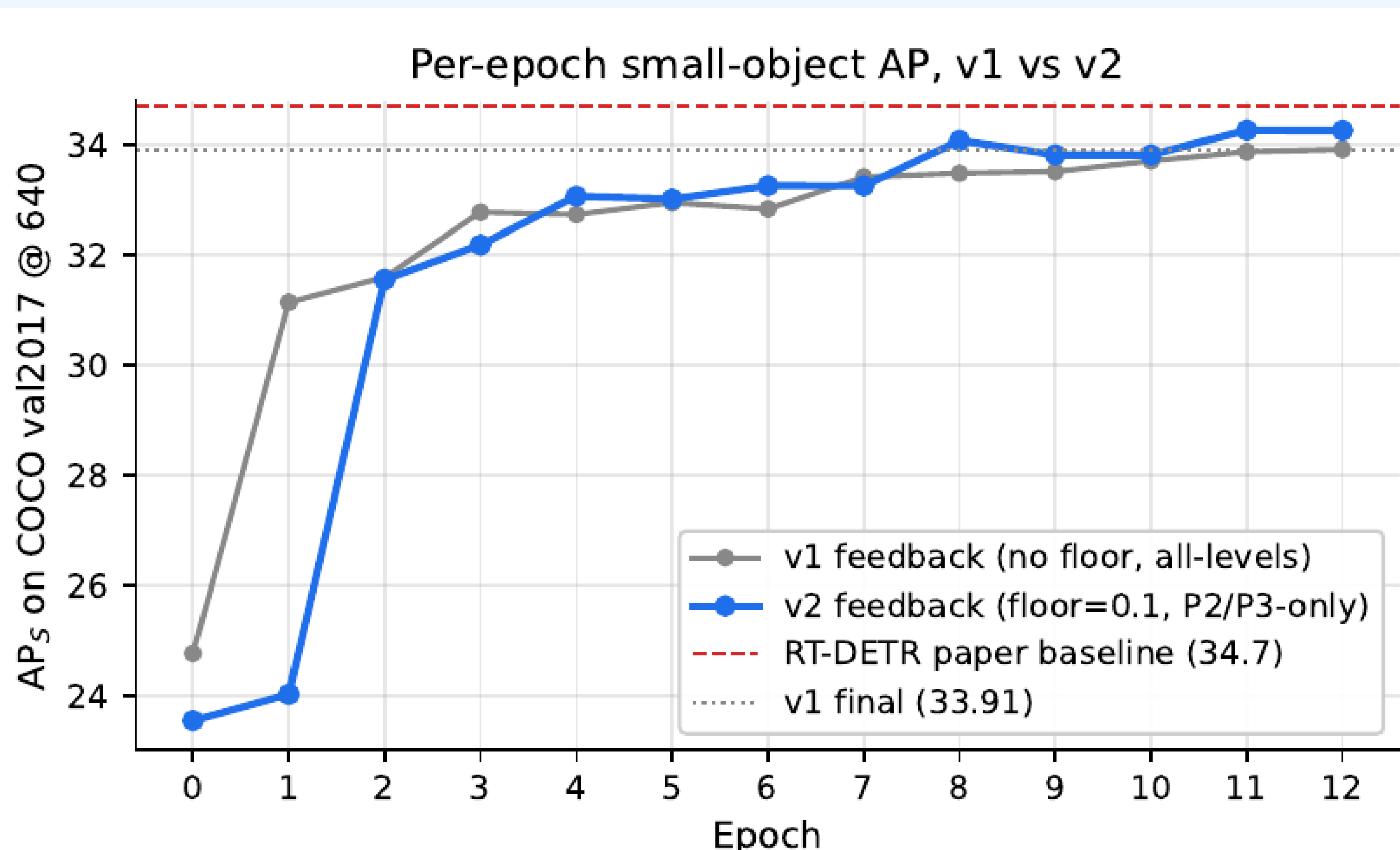
Two evaluations of identical weights, identical inputs — the only difference is whether the feedback module's forward call fires at inference. Any difference in metrics is therefore attributable to the feedback computation alone.

6. Results

Model	Mode	AP	AP _S	AP _M	AP _L
Feedback v1	ON	52.01	33.91	56.28	68.83
Feedback v1	OFF	51.92	33.91	56.18	68.63
v1 Δ causal	—	+0.09	0.00	+0.10	+0.20
Feedback v2	ON	52.10	34.25	56.33	68.82
Feedback v2	OFF	51.53	33.26	55.91	68.32
v2 Δ causal	—	+0.57	+0.99	+0.42	+0.50



7. Training Trajectory



v2 first crosses v1's eventual final 33.91 at **epoch 8** — four epochs early.

8. Limitations & Future Work

- Limitations.**
 - Single seed; multi-seed variance unmeasured.
 - 13-epoch finetune (vs 72-epoch from scratch).
 - Floor and mask not isolated — a 2x2 ablation would resolve.
 - $\approx 2\times$ slower (cost is the P2 level, not feedback).

Future work — recovering FPS.

- Train with P2, infer without.
 - Keep the gain; recover ≈ 53 FPS.
- Lighter P2 path (fewer channels / simpler projection).
- Knowledge distillation v2 $\rightarrow$ P3-P5 student.

An unconstrained sigmoid gate lets the optimizer silence a new mechanism — a floor closes that route.

The difference between a null and a +0.99 ablation. Built on RT-DETR (Lyu et al., CVPR 2024).

github.com/HatemSaadallah/feedback-augmented-rtdetr



scan for code & paper